

API Documentation — Photo Archive & Document Management System v1.0

ຮ້າງອີງ Book 1 §22.1 (Layer), §23 (API Standards), §54 (Error Codes). client ຮຽກ server ຜ່ານ google.script.run. Mutating actions ຜ່ານ Backend Route Guard (Code.gs api() → ROUTES → guard()); ຫຼື ຮຽກ Service function ໂດຍກົງທີ່ ບັງຄັບ CSRF+RBAC ພາຍໃນ.

1. ຮູບແບບການຮຽກ (Invocation)

```
// read ຜ່ານ router (guard: session + RBAC):
google.script.run.api({ token, action:"photos.list", payload:{...} })
```

```
// mutating (CSRF ບັງຄັບ) — ຮຽກ Service ໂດຍກົງ:
google.script.run.photoUpload(payload, token, csrf)
```

Auth model: login() → { token, user, csrf }. token = session (CacheService, idle 30m, rotation). csrf ຜູກ session; ທຸກ mutating ຕ້ອງສົ່ງ csrf.

2. Router Routes (Code.gs)

action	module/action	handler
photos.list	photos/view	listPhotos(payload)
orders.list	orders/view	listAll(Orders)
orders.print	orders/print	update Orders Status=Printing
documents.list	documents/view	listDocuments(payload)
documents.search	documents/view	searchDocuments(query,filters)
search.query	search/view	search(query,filters,opts)
reports.kpi	reports/view	getKpiReport(token)
dashboard.data	dashboard/view	getDashboardData(payload)

3. Service Functions

Auth (Auth.gs)

Function	Params	Returns
login(meta)	meta {ip,browser,device }	{token,user,csrf,ttl}
logout(token,meta)	token	true

guard(token,module,action,meta)	—	user throw E001/E005
getUiPermissions(token)	token	{role,modules}
requireCsrf(token,csrf)	—	session throw E005

Photo (PhotoService.gs)

Function	Params	Returns
photoUpload(payload,token,csrf,meta)	{name,mime,dataBase64,customerCode,category}	{photoId,thumbUrl,status} {duplicate}
photoBatchUpload(files,opts,token,csrf)	files[]	{success,failed,duplicates}
listPhotos(opt)	{customerCode,status}	[{photoId,thumbUrl,status}]
archivePhoto(photoId,token,csrf)	photoId	{status:Archived}

Document (DocumentService.gs)

Function	Params	Returns
documentUpload(payload,token,csrf,meta)	{title,category,name,mime,dataBase64,replaceDocId}	{docId,previewUrl,version}
listDocuments(opt)	{customerCode,category}	[brief]
searchDocuments(query,filters)	query,{category}	[brief]
getDocumentPreview(docId,token)	docId	{previewUrl}
getDocumentVersions(docId)	docId	[versions]

Search (SearchService.gs)

Function	Params	Returns
search(query,filters,opts)	filters{refType,customerCode,category,status,fileType,dateFrom,dateTo}	{results,total,tookMs}
indexRef(refType,refId)	—	(incremental index)
getRecentSearches() / saveSearchFilter(name,filters)	—	[q] / map

Report (ReportService.gs)

Function	Params	Returns
getKpiReport(token)	token	{customers,photos,documents,orders,storageMB}
getTrendReport(period,token)	daily monthly yearly	[{period,uploads,orders}]
getTopCategories(token)	token	{photoCategories,docCategories,jobTypes}
getStorageAnalytics(token)	token	{usedMB,limitMB,percent,warn}
exportReport(type,format,token)	kpi trend top storage, csv xlsx pdf	{filename,mime,content dataBase64}

Settings (SettingsService.gs) — Admin

Function	Params	Returns
getSettings/ updateSetting(key,value,token)	—	{...}
getFeatureToggles/ setFeatureToggle(name,on,token)	—	{...}
listUsers/ createUserRec/ updateUserRec	—	[...]
backupNow/listBackups/ restoreFromBackup(id,token)	—	{...}
getAuditLogs(opt,token)	{eventType,userId,date,page}	{rows,total}
getMetricsSummary(token)	token	{metrics,cacheHitRatePct,benchmarks}

4. Request / Response (JSON)

```
// Request
{ "action": "search.query",
  "token": "<uuid>",
  "payload": { "query": "CUS-000001", "filters": { "refType": "photo" }, "opts":
    { "page": 1, "size": 20 } } }
```

```
// Response
{ "results":[ { "refType":"photo","refId":"PHOTO-000012","name":"...","status":"Ready"} ],
  "total":42, "tookMs":180, "page":1, "size":20 }
```

5. Error Codes (§54)

Code	ຄວາມໝາຍ
E001	Login Failed / Session ໝົດ
E002	Upload Failed
E003	Drive Quota
E004	Not Found (Folder/Sheet/record/action)
E005	Permission Denied / CSRF / Rate limit
E006	Invalid File Type
E007	File Too Large
E_VALIDATION	Input ບໍ່ຖືກ (enum/required)

△ ຮູບແບບ Error: "Exxx: <ຂໍ້ຄວາມ i18n> — <extra>". ຫ້າມສະແດງ stack trace ໃຫ້ User (§13); ເກັບໃນ AuditLogs.